

**Fundamental Principle of COUNTING**

"If a thing can be done in m different ways and another thing can be done in n different ways, then the two things can be done in $m \times n$ different ways."

Number of ways, $N = m \cdot n$

Example: You have 3 ways to wear t-shirt and 2 ways to wear shorts.



The number of ways N is $N = 3 \cdot 2 = 6$

**Permutation**

Permutation is as an ordered arrangement of a finite number of elements, either all of the available n elements or of a part of them. The permutations that contains exactly the same elements but not in the same order are regarded as different.

The number of permutations of n different things

taken r at a time is ${}_nP_r = \frac{n!}{(n-r)!}$

If taken all at once, ${}_nP_n = n!$

Inversion

If two elements in a permutation of distinct elements are in reverse order relative to their normal or natural order, they constitute an **inversion**. A permutation is said to be **even** if it contains an even number of inversions; it is **odd** if the number of inversions is odd.

The number of transpositions that are required to return a sequence of elements to their natural order is even or odd according to the number of inversions in the arrangement.

Example:

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In the permutation (C, E, D, B, A)

Element C precedes B and A, therefore has 2 inversions.

Element E precedes D, B and A, therefore has 3 inversions.

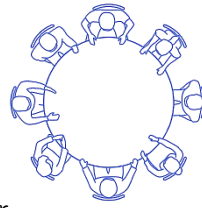
Element D precedes B and A, therefore has 2 inversions

Element B precedes A, therefore has 1 inversion.

The total inversion is $2 + 3 + 2 + 1 = 8$. So permutation (C, E, D, B, A) is an even permutation.

Cyclic Permutation

Cyclic permutation is the shifting of an entire order of elements one or more steps forward or backward – the first element taking the position of the last, or vice versa, without changing the order of the elements in the sequence.



$${}_nP_n = (n-1)!$$

Example:

How many ways can 6 people be seated in a roundtable?

$${}_6P_6 = (6-1)! = 120 \text{ ways}$$

Permutation with Identical Elements

The number of permutations is reduced when a collection contains **identical elements**. The number of permutation n objects, p of one type, q of another, s of another, etc., is given by:

$$P = \frac{n!}{p! q! s!}$$

Example 1: How many permutations can be made with 5 bananas, 4 apples and 3 mangoes?



$$P = \frac{n!}{p! q! s!} = \frac{(5+4+3)!}{5!4!3!} = 27,720$$

Example 2: Find the largest number of permutations of the letters in MATHEMATICS?

Solution:

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The word MATHEMATICS has 11 letters. The letter M occurs 2 times, letter A occurs 2 times, letter T occurs 2 times while the rest occurs once.

$${}_11P_{11} = \frac{11!}{(11-11)! 2! 2! 2! 1! 1! 1! 1! 1! 1! 1!}$$

$${}_11P_{11} = 4989600$$



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Assortment

Assortment refers to a group of objects selected from a larger group in such a way that an object can be used more than once.

The number of assortments is expressed mathematically as:

$$\text{Assortments} = (\text{No. of choices for 1st position}) \times (\text{no. of choices for the 2nd position}) \times (\text{no. of choices for the 3rd position}) \times \dots \times (\text{no. of choices for last position})$$

Combination

Combination is an arrangement of the selection of objects regardless of the order. The number of combinations of n different things taken r at a time is:

$${}_nC_r = \frac{n!}{(n-r)!r!}$$

If taken all at once, ${}_nC_n = 1$

Relation between Permutation and Combination

The relation between permutation of n thing taken r at a time to the combination of n things taken r at a time is expressed mathematically as

$$\text{follows: } {}_nC_r = \frac{{}_nP_r}{r!}$$

Probability

Probability is simple defined as the numerical assessment of likelihood. It is expressed as a number between 0 and 1, where 0 means an event is impossible while 1 means it is surely or certainly to occur or happen. The total probability of favorable and unfavorable outcomes is 1. If an experiment has a set of distinct outcomes, each of which is equally likely to happen, then the probability of an event E , is the ratio of the number of outcomes to the total number of possible outcomes.

$$P_E = \frac{\text{no. of outcomes}}{\text{total outcomes}}$$

Probability that it is **not E** is: $P_{\text{not } E} = 1 - P_E$

One of the earliest mathematical studies on probability was "On Casting the Die", written by the 16th century Italian mathematician and physician **Gerolamo Cardano** (1501 – 1576). Cardano defined probability as "the number of favorable outcomes divided by the number of possible outcomes". Cardano was regarded as "the father of the theory of probability".



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